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Programme: PCDFIR - Cohort 11 (C11)

Course code: CIP-B101

Course: Basic Computer Skills for Digital Forensics

Assignment title: Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

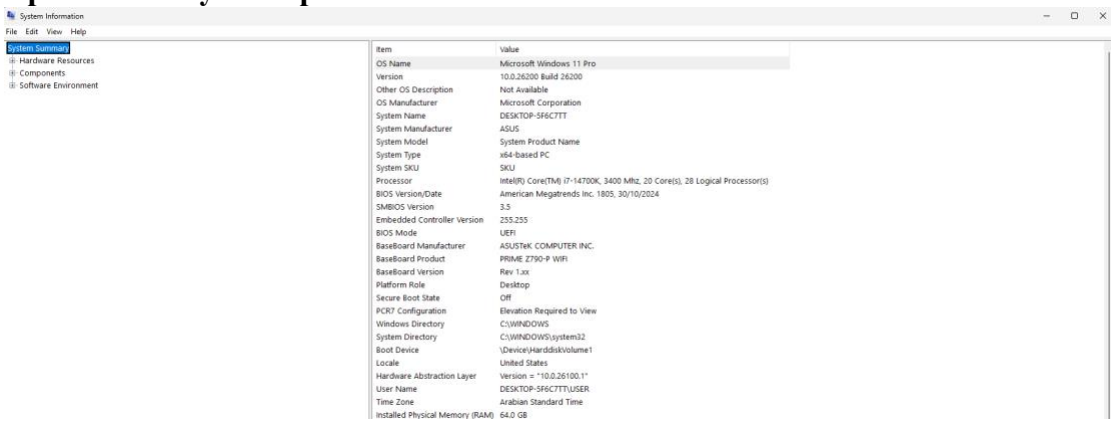
Platform used: ICDFA Kali Linux, PowerShell, Command Prompt and Windows 11 Desktop.

Date: 6th November 2026

6. Lab Scenario

Part A - System Identification and Evidence Layer Mapping

A1. Capture basic system specification

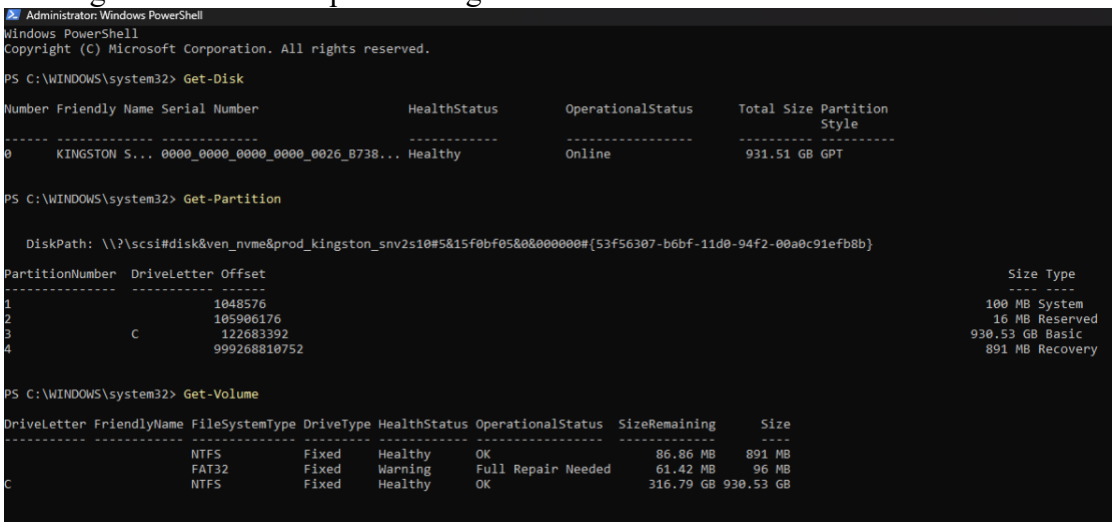



```
C:\Windows\System32>systeminfo

Host Name:                 DESKTOP-5F6C7TT
OS Name:                   Microsoft Windows 11 Pro
OS Version:                10.0.26200 N/A Build 26200
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          USER
Registered Organization:   N/A
Product ID:                00330-53573-12105-AAOEM
Original Install Date:     19/11/2024, 7:22:24 AM
System Boot Time:          28/10/2026, 10:44:52 AM
System Manufacturer:       ASUS
System Model:              System Product Name
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 183 Stepping 1 GenuineIntel ~3400 Mhz
BIOS Version:              American Megatrends Inc. 1805, 30/10/2024
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:             en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+04:00) Abu Dhabi, Muscat
Total Physical Memory:     65,253 MB
Available Physical Memory: 48,385 MB
Virtual Memory: Max Size:  69,349 MB
Virtual Memory: Available: 46,855 MB
Virtual Memory: In Use:    22,494 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\DESKTOP-5F6C7TT
Hotfix(s):                 4 Hotfix(s) Installed.
                           [01]: KB5127690
                           [02]: KB5054156
                           [03]: KB5127648
                           [04]: KB5127647
Network Card(s):           6 NIC(s) Installed.
                           [01]: Realtek Gaming 2.5GbE Family Controller
                                 Connection Name: Ethernet
                                 Status:          Media disconnected
                           [02]: Intel(R) Wi-Fi 6 AX201 160MHz
                                 Connection Name: Wi-Fi
                                 DHCP Enabled:    Yes
                                 DHCP Server:     192.168.1.1
                                 IP address(es)
                                 [01]: 192.168.1.222
                                 [02]: fe80::88cd:8951:41b6:66c0
                           [03]: VirtualBox Host-Only Ethernet Adapter
                                 Connection Name: Ethernet 2
                                 DHCP Enabled:    No
                                 IP address(es)
                                 [01]: 192.168.56.1
                                 [02]: fe80::c0f1:15e4:9561:221a
                           [04]: VMware Virtual Ethernet Adapter for VMnet1
                                 Connection Name: VMware Network Adapter VMnet1
                                 DHCP Enabled:    Yes
                                 DHCP Server:     192.168.148.254
                                 IP address(es)
```

A2. Capture storage information

– Using PowerShell to capture storage information



A3. Map evidence sources to system layers

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
Network layer	– IP addresses, – DNS requests, – Websites visited – Network connection records	In firewall logs, router logs, DNS logs or in packet captures.	It helps to identify the communication between devices and any external systems.
User layer	– Login attempts – Files copied to a USB – Accessed file/folders	In the user activity records, USB usage history and in the access records.	It verifies what actions were done by any user.
Application layer	– Downloaded attachments – Email headers – Application settings	In the email client data, application configuration files and even in the local application storage.	It helps to identify how applications were used and what information they processed.
Disk layer	– File creation dates – File permissions – Directory entries	In the file's system metadata, disk partitions and storage volumes.	It helps to determine how files were stored, changed or removed.
Memory/ Process layer	– Active processes – Loaded modules – Encryption keys	RAM captures, process and in the module listings.	It helps to identify what was running on the system at the time of examination.
Operating System (OS) layer	– System startup records, software installation events, account creation records	Windows Event Viewer, Windows Registry, Linux system logs.	Helps establish system activity, configuration changes and any user account related activities.

Why may evidence exist in more than one layer?

A single user action can produce records in multiple parts of a computer hence leading to existence of evidence in different system layers. For instance; A user downloads an email attachment the network layer will record the connection, email and its attachments will be stored in the application layer, the downloaded file will be stored in the disk and records of that activity will be found on the operating system. Thus evidence examination from different layers helps an investigator to compare information and build an accurate timeline to prove what happened.

Part B - Program, Process and Thread Investigation

B1. Observe running processes

- Screenshot showing the running processes on the machine

```
PS C:\WINDOWS\system32> Get-Process | Select-Object -First 20 Name,Id,CPU,WorkingSet

Name                              Id          CPU WorkingSet
----                              --          --- ----------
Aac3572DramHal_x86              8620     0.046875   11431936
Aac3572MbHal_x86               15656    10.796875   11796480
Aac3572MbHal_x86               17192 21755.140625   14544896
AacAmbientLighting             14452     3.734375   36651008
AacKingstonDramHal_x64         10288    19.515625   11501568
AacKingstonDramHal_x86          6212    11.171875   11214848
AcPowerNotification            12348   126.453125    9916416
AggregatorHost                 10028    18.09375    20844544
AppleMobileDeviceService        6336   261.359375   15278080
ApplicationFrameHost           23804     4.671875   52076544
ArmouryCrate.Service            6380      71.3125   54128640
ArmouryCrate.UserSessionHelper 12124     2878.125   80093184
ArmourySocketServer            12336        0.625    2654208
ArmourySwAgent                 19124     0.453125    2326528
asus_framework                 13324    526.65625   17539072
asus_framework                 17740     667.5625   10846208
asus_framework                 17916      23.9375    6361088
asus_framework                 17936       45.375   10461184
asus_framework                 19180        0.375    6172672
asus_framework                 19292       12.875    6496256
```

B2. Explain the difference

A program is a set of instruction stored on a hard drive such as ***.exe file*** or ***an application*** installed in the computer, while **a process** is an active running instance of a program loaded in the computer's Random Access Memory(RAM) and **a thread** is the smallest unit of execution that an operating system can schedule and send to the CPU.

Explain why process information may be useful during malware analysis or live forensic triage.

It can be used by an investigator to identify all programs that are running at that moment and even facilitate detection of any suspicious activity. The process names, IDs (PIDs), locations of files , CPU and memory usage and network connections can help an investigator to identify unusual processes that may indicate malware and show if that malware is active , any processes it

has connected with and whether it is communicating with external systems. All this information is used to collect relevant evidence, get familiar with the current computer state before it shuts down or the process stops running.

Part C - Disk Sector and Disk Size Calculation

Key formula
Disk or partition size = number of sectors x bytes per sector. To convert bytes to GB, divide by 1024 x 1024 x 1024.

C1. Reproduce the slide calculation
Use the values from the slide to calculate the total disk size: sectors per track = 400, heads = 12, cylinders = 17000, and bytes per sector = 512.

Item	Formula / Value	Answer
Total sectors	400 x 12 x 17000	81,600,000 sectors
Total bytes	81,600,000 x 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 / 1024	40,800,000 KB
Megabytes	40,800,000 / 1024	39,843.75 MB
Gigabytes	39,843.75 / 1024	38.91 GB

Given values:

- Sectors per track = 400
- Number of heads = 12
- Number of cylinders = 17,000
- Bytes per sector = 512

Calculating total sectors;
Formula:
Total Sectors = Sectors per Track × Number of Heads × Number of Cylinders
=400 × 12 × 17,000
=**81,600,000 sectors**

Calculating the total bytes
Formula:
Total Bytes = Total Sectors × Bytes per Sector
= 81,600,000 × 512
=**41,779,200,000 bytes**

Converting bytes to kilobytes (KB)
KB = (Total Bytes)/1024

= (41,779,200,000)/1024
=**40,800,000 KB**

Converting kilobytes to megabytes (MB)
MB = $(\text{KB})/1024$
= $(40{,}800{,}000)/1024$
=**39,843.7 MB**

Converting megabytes to gigabytes (GB)
GB = $(\text{MB})/1024$
= $(9{,}843.7)/1024$
=**38.9 GB**

Answer:
Total calculated disk capacity = 38.91 GB

C2. Calculate your own disk or partition size

- Screenshots are in the Appendix
- Calculations of partition size

Formula:

$$Partition\ Size\ (GB) = (Sector\ Count\ \times 512)/(1024 \times 1024 \times 1024)$$

Partition 1: System (FAT32)

Sector count = 204,800

Then; Partition Size (GB) $= (204{,}800 \times 512)/(1024 \times 1024 \times 1024)$
= 0.0976 GiB

Partition 2 : **Basic (C:) (NTFS)**

Sector count = 1,951,457,280
Partition Size (GB) $= (1{,}951{,}457{,}280 \times 512)/(1024 \times 1024 \times 1024)$

= 930.5273 GiB.

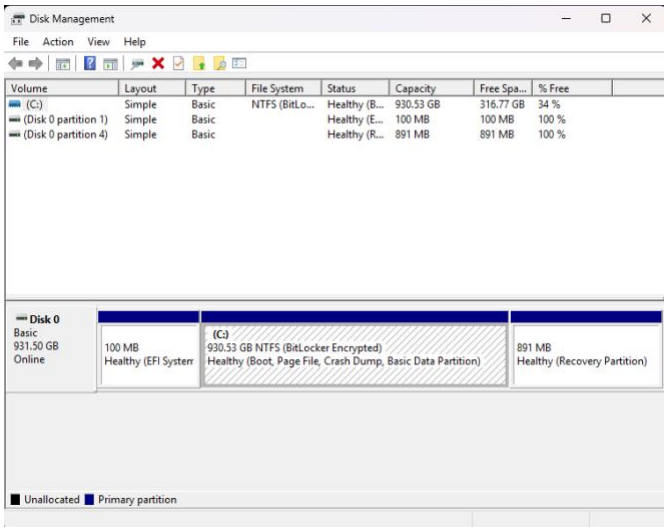
Partition 3 : Recovery (NTFS)

Sector count = 1,824,768
Partition Size (GB) $= (1{,}824{,}768 \times 512)/(1024 \times 1024 \times 1024)$

= 0.87011 GiB.

Part D - Safe Partition Table Inspection

D1. Windows method - Disk Management, PowerShell and Disk Part




```
PS C:\WINDOWS\system32> Get-Disk

Number Friendly Name                                                                                Serial Number                    HealthStatus         OperationalStatus      Total Size Partition
                                                                                                                                                                                   Style
------ -------------                                                                                ------------                     ------------         -----------------      ---------- ----------
0      KINGSTON SNV2S1000G                                                                          0000_0000_0000_0000_0026_B738... Healthy              Online                  931.51 GB GPT


PS C:\WINDOWS\system32> Get-Partition


   DiskPath: \\?\scsi#disk&ven_nvme&prod_kingston_snv2s10#5&15f0bf05&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                                                                                    Size Type
---------------  ----------- ------                                                                                                    ---- ----
1                            1048576                                                                                                 100 MB System
2                            105906176                                                                                                16 MB Reserved
3                C           122683392                                                                                               930.53 GB Basic
4                            999268810752                                                                                            891 MB Recovery


PS C:\WINDOWS\system32> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus  SizeRemaining      Size
----------- ------------ -------------- --------- ------------ -----------------  -------------      ----
                         NTFS           Fixed     Healthy      OK                       86.86 MB    891 MB
                         FAT32          Fixed     Warning      Full Repair Needed       61.42 MB     96 MB
C                        NTFS           Fixed     Healthy      OK                      316.76 GB 930.53 GB
```

```
PS C:\WINDOWS\system32> diskpart

Microsoft DiskPart version 10.0.26100.1150

Copyright (C) Microsoft Corporation.
On computer: DESKTOP-5F6C7TT

DISKPART> list disk

  Disk ###  Status         Size     Free     Dyn  Gpt
  --------  -------------  -------  -------  ---  ---
  Disk 0    Online          931 GB  1024 KB        *

DISKPART> select disk 0

Disk 0 is now the selected disk.

DISKPART> list partition

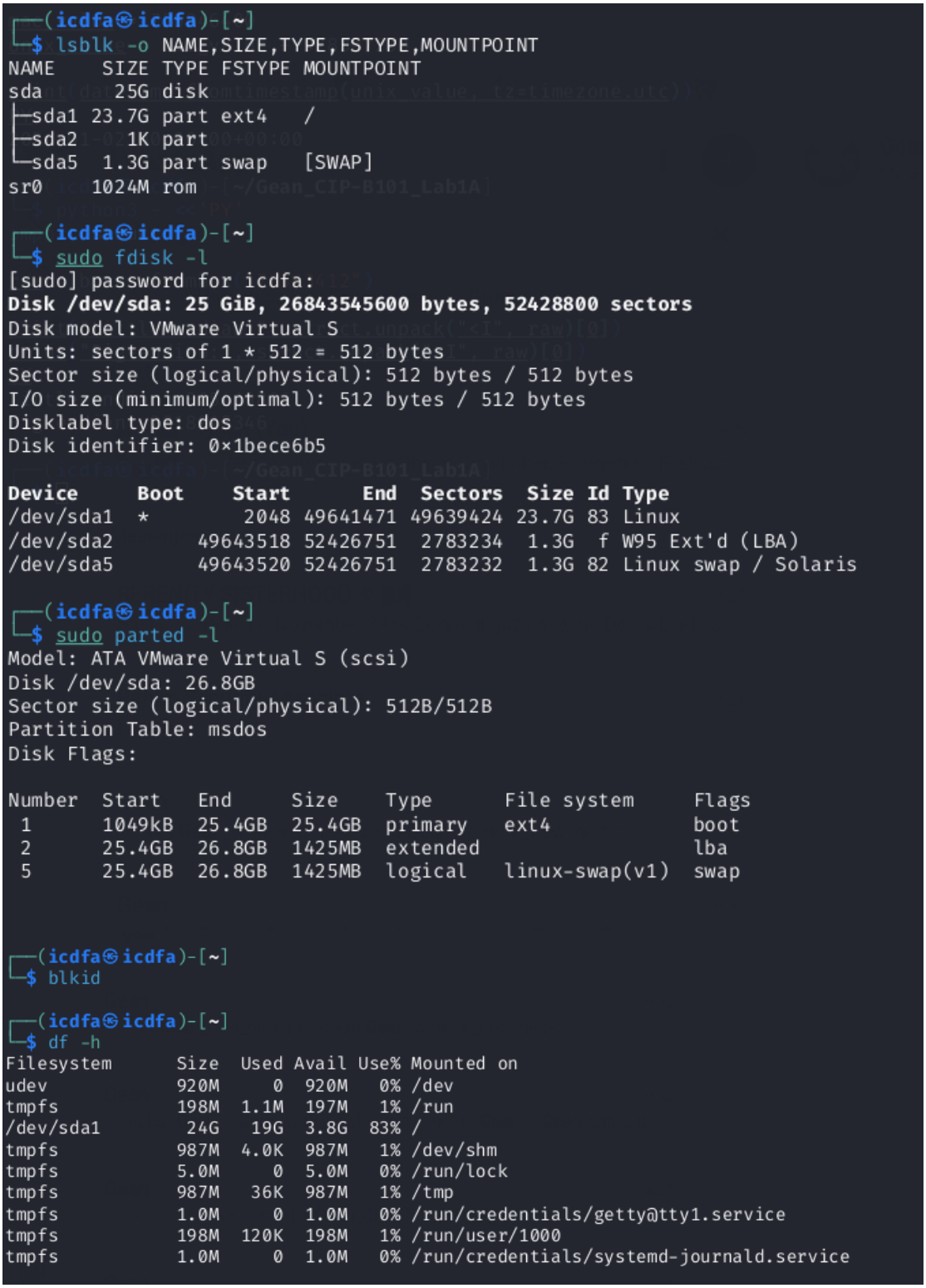
  Partition ###  Type              Size     Offset
  -------------  ----------------  -------  -------
  Partition 1    System             100 MB  1024 KB
  Partition 2    Reserved            16 MB   101 MB
  Partition 3    Primary            930 GB   117 MB
  Partition 4    Recovery           891 MB   930 GB

DISKPART> exit

Leaving DiskPart...
PS C:\WINDOWS\system32>
```

D2. Linux/Kali method - read-only inspection

- Screenshot showing disk, partition, size and file system information



```
┌──(icdfa㉿icdfa)-[~]
└─$ lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
NAME    SIZE TYPE FSTYPE MOUNTPOINT
sda      25G disk
├─sda1 23.7G part ext4   /
├─sda2    1K part
└─sda5  1.3G part swap   [SWAP]
sr0    1024M rom

┌──(icdfa㉿icdfa)-[~]
└─$ sudo fdisk -l
[sudo] password for icdfa:
Disk /dev/sda: 25 GiB, 26843545600 bytes, 52428800 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0×1bece6b5

Device     Boot    Start      End  Sectors  Size Id Type
/dev/sda1  *        2048 49641471 49639424 23.7G 83 Linux
/dev/sda2       49643518 52426751  2783234  1.3G  f W95 Ext'd (LBA)
/dev/sda5       49643520 52426751  2783232  1.3G 82 Linux swap / Solaris

┌──(icdfa㉿icdfa)-[~]
└─$ sudo parted -l
Model: ATA VMware Virtual S (scsi)
Disk /dev/sda: 26.8GB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:

Number  Start   End     Size    Type      File system     Flags
 1      1049kB  25.4GB  25.4GB  primary   ext4            boot
 2      25.4GB  26.8GB  1425MB  extended                  lba
 5      25.4GB  26.8GB  1425MB  logical   linux-swap(v1)  swap


┌──(icdfa㉿icdfa)-[~]
└─$ blkid

┌──(icdfa㉿icdfa)-[~]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            920M     0  920M   0% /dev
tmpfs           198M  1.1M  197M   1% /run
/dev/sda1        24G   19G  3.8G  83% /
tmpfs           987M  4.0K  987M   1% /dev/shm
tmpfs           5.0M     0  5.0M   0% /run/lock
tmpfs           987M   36K  987M   1% /tmp
tmpfs           1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           198M  120K  198M   1% /run/user/1000
tmpfs           1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
```

D3. Optional WinHex or forensic viewer method

- Screenshot showing the disk am using to be GPT.

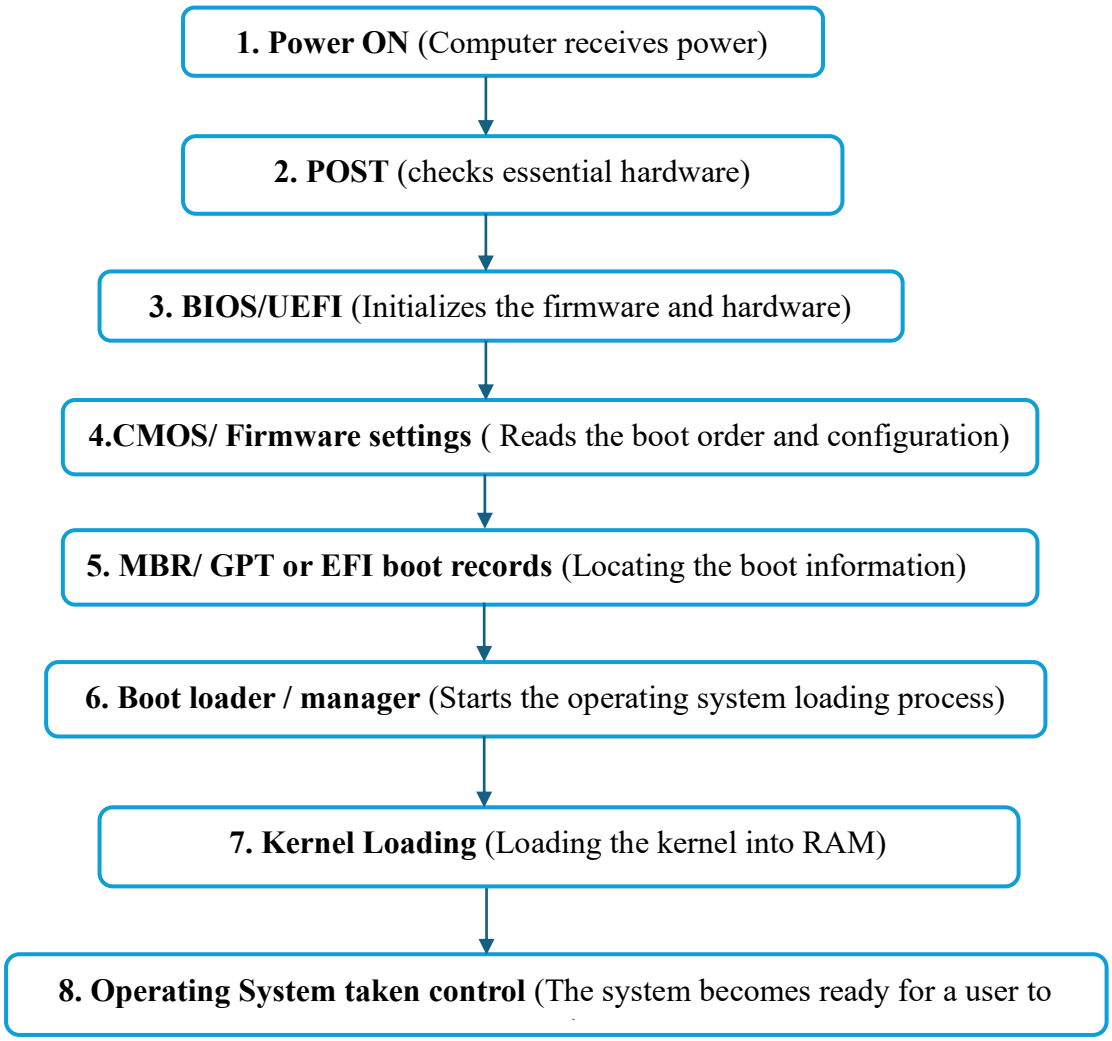
```
PS C:\WINDOWS\system32>  Get-Disk | Select-Object Number, FriendlyName, PartitionStyle, LogicalSectorSize, Size


Number            : 0
FriendlyName      : KINGSTON SNV2S1000G
PartitionStyle    : GPT
LogicalSectorSize : 512
Size              : 1000204886016
```

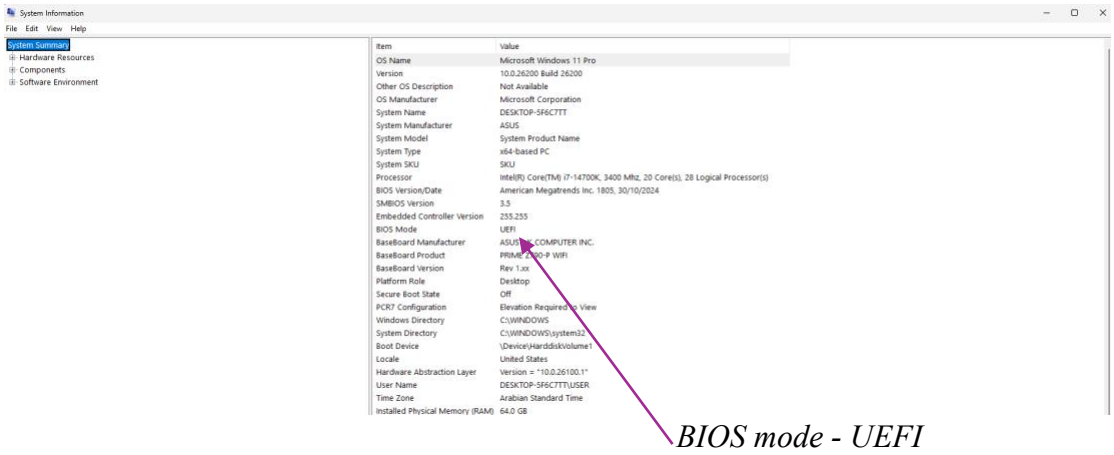
Part E - MBR, GPT and Boot Process Explanation

E1. Explain the boot process

- A simple flow diagram on how a PC boots.



- My computer uses the **UEFI (Unified Extensible Firmware Interface)** boot mode. This is supported by my system's GPT partition style and the presence of a FAT32 System partition that is used for EFI boot files. And the BIOS Mode information in System Information as well confirms that the computer uses UEFI as shown in the screenshot below;



BIOS mode - UEFI

E2. Comparison between MBR and GPT

Feature	MBR (Master Boot Record)	GPT(GUID Partition Table)
Partition Limit	Can support maximum up to 4 primary partitions.	Can support up to 128 partitions in a standard Windows usage.
Maximum disk size	Can support disks up to approximately 2 TB with a 512-byte sectors.	Can support much larger disks up to 9.4 ZB with a 512-byte sectors.
Boot mode	Mostly used in Legacy BIOS	Normally used in UEFI BIOS mode.
Partition information	It stores partition information in the first sector of the disk.	It stores partition information in a GPT header and partition entry array.
Backup	Can not provide a separate backup partition table.	Keeps a backup GPT header and the partition entry array at the end of the disk.
Data protection	It has limited protection against partition table corruption.	Uses CRC checksums to detect corruption in its GPT structures.
Compatibility	Mostly supported by older computers and operating systems.	Supported by the modern computers and operating systems.

Part F - File System Identification and Interpretation

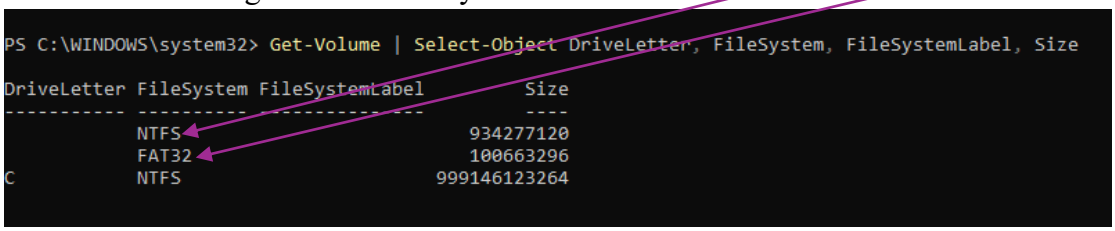
- Identify the file system used by at least two volumes or partitions on your system.

Answer:

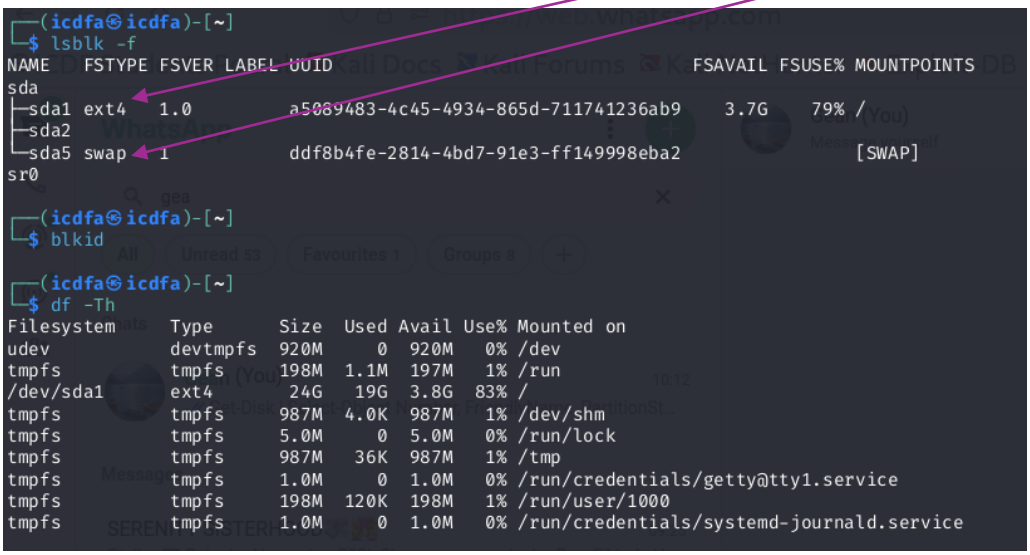
My computer (Windows System) uses the FAT32 and NTFS file systems as shown in screenshot below;

- **FAT32:** Is used by the EFI System Partition to store boot files needed during startup.
- **NTFS:** Used by the main Windows partition (C:) and the recovery partition to organize and manage files.

- Screenshot showing the Windows system volumes are NTFS and FAT32.



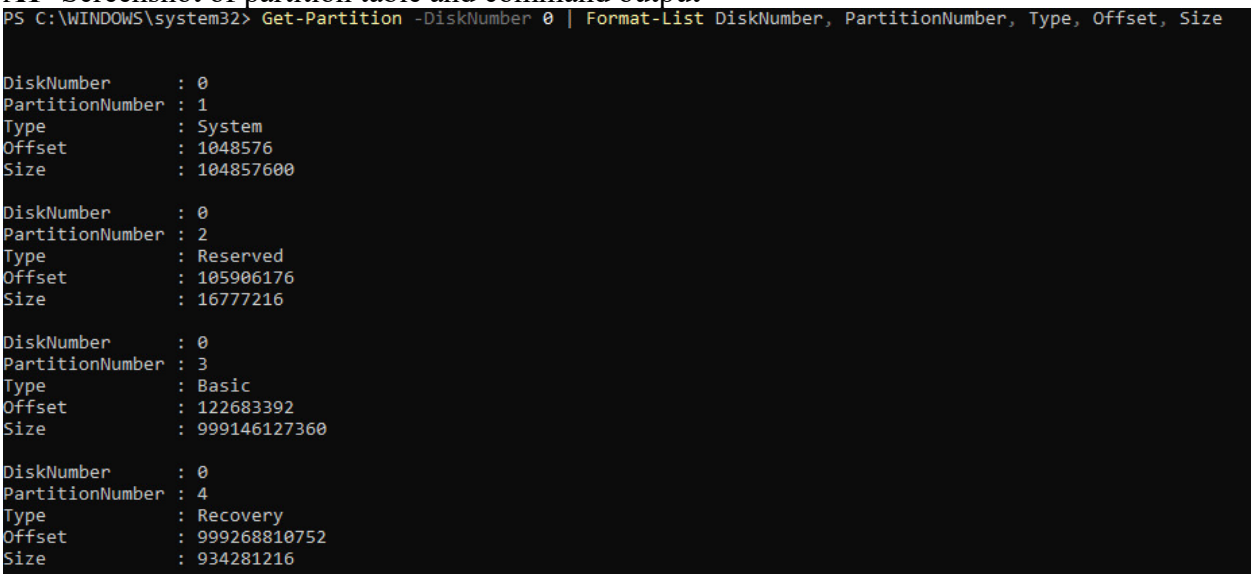
- Screenshot showing the Linux system partitions used are ext4 and swap.



Part G - Final Practical Assignment

Task	Evidence	Answer
Identify partition 1	Screenshot of partition table and command output	**A1**
Record partition 1 sector count	Start sector, end sector or sector count	**A2**
Calculate partition 1 size	Formula and result in GB	**A3**
Identify partition 2	Screenshot of partition table and command output	**A1**
Record partition 2 sector count	Start sector, end sector or sector count	**A4**
Calculate partition 2 size	Formula and result in GB	**A5**
Compare with OS-reported size	Disk Management, Get-Volume, lsblk, or df output	**A6**
Explain difference if any	Short explanation of rounding, reserved space, file system overhead, or GiB/GB differences	**A7**

- **A1-** Screenshot of partition table and command output



- **A2**- Record Partition 1 sector count
  – Calculating the starting sector;

$$\text{Start Sector} = Offset \ / \ (Sector\ Size)$$
$$= 1048576 \ / \ 512$$
$$= \mathbf{2048}$$

  - Calculating the number of sectors;

$$\text{Sector count} = Partition\ Size \ / \ (Sector\ Size)$$
$$= 104857600 \ / \ 512$$
$$= \mathbf{204800\ sectors}$$

  - Calculating the ending sector;

$$\text{End Sector} = \text{Start Sector} + \text{Sector Count} - 1$$
$$= 2048 + 204800 - 1$$
$$= \mathbf{206847}$$

- **A3 -** Calculate Partition 1 size
  - Formula;

$$\text{Partition Size (GB)} = Sector\ Count \times 512 \ / \ 1024^3$$
$$= 204800 \times 512 \ / \ 1024^3$$
$$= \mathbf{0.097\ GiB}$$

- **A4 -** Record Partition 2 sector count
  – Calculating the starting sector;

$$\text{Start Sector} = Offset \ / \ (Sector\ Size)$$
$$= 105906176 \ / \ 512$$
$$= \mathbf{206848}$$

  - Calculating the number of sectors;

$$\text{Sector count} = Partition\ Size \ / \ (Sector\ Size)$$
$$= 16777216 \ / \ 512$$
$$= \mathbf{32768\ sectors}$$

  - Calculating the ending sector;

$$\text{End Sector} = \text{Start Sector} + \text{Sector Count} - 1$$
$$= 2206848 + 32768 - 1$$
$$= \mathbf{239615}$$

- **A5 -** Calculate Partition 2 size
  Formula;

$$\text{Partition Size (GB)} = Sector\ Count \times 512 \ / \ 1024^3$$
$$= 32768 \times 512 \ / \ 1024^3$$
$$= \mathbf{0.0157\ GiB}$$

- **A6-** Compare with OS-reported size

```
PS C:\WINDOWS\system32> Get-Volume | Select-Object DriveLetter, FileSystem, FileSystemLabel, Size

DriveLetter FileSystem FileSystemLabel         Size
----------- ---------- ---------------         ----
            NTFS                          934277120
            FAT32                         100663296
C           NTFS                        999146123264
```

Partition 1 has a calculated size of **100 MiB** while Windows shows its FAT32 volume as **96 MiB**. This difference is because some of the space is used by the file system to manage files and store its information thus affecting the usable volume size.

Partition 2 has a calculated size of **16 MiB**, but Windows does not show its size in "*Get-Volume*" because it is a reserved partition and has no file system or drive volume therefore, there is no OS-reported volume size to compare against its calculated partition size.

- **A7** : Short explanation of rounding, reserved space, file system overhead, or GiB/GB differences

My examination of Disk 0 showed two partitions. Partition 1 is a System partition with a calculated size of 0.10 GiB, while Partition 2 is a Reserved partition with a calculated size of 0.02 GiB. The calculations used the number of sectors and a sector size of 512 bytes.

When compared with Windows, Partition 1 had a reported volume size of 96 MiB instead of 100 MiB. This difference is because some space is used by the FAT32 file system to store its management information. Partition 2 had no reported volume size because it is reserved for Windows and does not have a file system.

Conclusion:

From this lab1B, I learned how to examine disk structures, identify partitions, understand file systems and to calculate storage sizes using sector information. I also learned how computers boot and how digital evidence can be found across different system layers. These skills are important in digital forensics because they will help me as a forensic investigator to understand how a computer stores and processes data, preserve the evidence and accurately reconstruct events during an investigation.